COMPOSING " IN TRANSIT"

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ABSTRACT

"In Transit" is a fixed form electro acoustic piece. It combines of a number of short improvisations performed on an instrument created by the composer using the ChucK programming environment. It enables the performer to create sequences of notes that display the characteristics of various probability distributions. The performer has macro level control of the output by adjusting parameters that control the algorithms, in real time. The improvisations are imported into a Digital Audio Workstation where they are edited and processed with various signal processors before being assembled to create the finished piece. The assembly makes use of a "bottom up" approach. There is no clear idea as to how the piece will be, the final form emerges from the way the composer combines the improvisations. The composer works largely intuitively, making use of high-level concepts such as progression, continuity and contrast as a framework to help make compositional decisions.

1. INTRODUCTION

"In Transit" is a fixed form stereo playback electro acoustic piece. It was created using a combination of ChucK, and Logic Pro. All sounds are produced with physical modeling synthesizers, but the timbres have been significantly transformed by a variety of signal processing techniques. The piece is predominantly organized around timbre and texture, and (mostly) does not exhibit a clearly articulated rhythmic pulse. A characteristic feature of the piece is the use of diffuse clouds of notes that fade in and out, which was the only element of the piece that was decided upon when the composition of the piece was begun. The overall form grew out of the composition process, but effort was made to make sure the piece had a sense of progression, as if the listener was on a leisurely journey, hence the name "In Transit".

The impetus for creating this piece came from a demonstration of the "Probability Jammer" in a music class. Each student was running "Probability Jammer" on his or her laptop and they all played with it through the laptop speakers. The class was divided up into groups, and students were directed to adjust the parameters in particular ways, "conducted" by the composer. This created an interesting cloud of notes in the room that changed as the students performed. This effect was the only preconceived characteristic that the composer set out to achieve in the final form of this piece.

2. COMPOSITIONAL APPROACH

"In Transit" is part of a collection of pieces that are created using a "bottom up" compositional methodology that attempts to apply some of the approaches the composer has successfully used in popular music to an electro acoustic context. In particular, there is the intention to include an element of improvised performance as the way of creating musical material, which is then further refined, out of real time. The piece is constructed in three distinct phases.

1. Create Real Time Improvisations
2. Apply Processing and Editing
3. Assembly/Mix

To use an analogy of a child building a plastic model kit (such as a plane/car/ship), the first phase involves creating the basic components (done by the model manufacturer), the second phase would be painting the parts, and the third phase, assembly. The big difference between the model building process and the compositional process is that the final form of that model is known from the beginning, while in this composition, the final form emerges in the assembly process, and is not apparent until the end. In that respect, it is like taking a collection of Lego bricks, and, without any preconceptions about what to make, putting pieces together randomly for a while, then assessing the situation to see if the assembly suggests any particular thing, then modifying this to refine it.

2.1. Phase 1 – Real Time Improvisations

In this piece, the basic building blocks were fourteen short solo (unaccompanied) improvisations performed on an instrument that the composer created with the ChucK programming language, called "Probability Jammer". This instrument makes use of two probability distribution generators to control the pitch and duration of notes that are produced using one of the STK physical modelling

synthesizers [1]. All of the STK instruments supplied with ChucK are available for use with the instrument. "Probability Jammer" has a very simple user interface, shown in figure 1, implemented using the MAUI widgets, part of the MiniAudicle ChucK development environment. Each onscreen fader can also be controlled via a MIDI fader box, which effectively allows all parameters to be altered simultaneously.

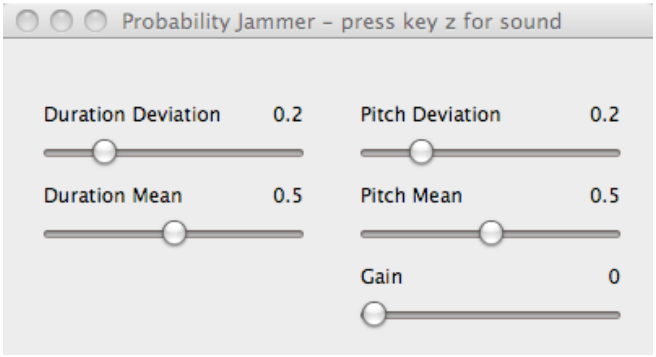


Figure 1. "Probability Jammer" user interface.

An example of the output of one of the probability generator functions (the Gaussian) with different settings is shown in figures 2, 3 and 4. The left hand window is a plot of the output produced when the generator is run 100,000 times without changing the parameters. The right hand window displays the corresponding (normalized) probability distribution function, and the relationship between changes in the parameter settings and the output produced is quite apparent.

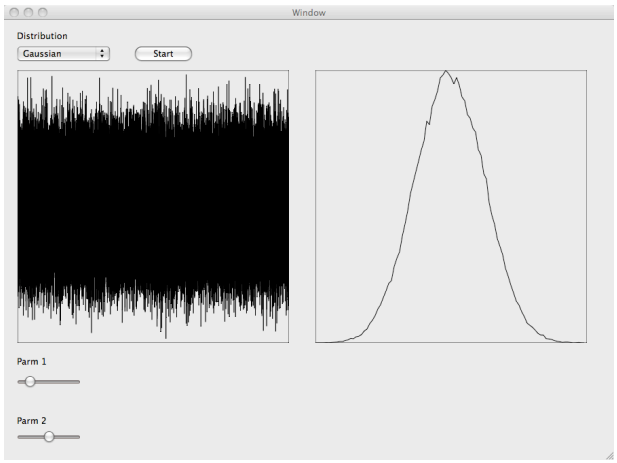


Figure 2. Output of the Gaussian generator, with its probability distribution for 100,000 samples. The mean is 0.5 and deviation is 0.2.

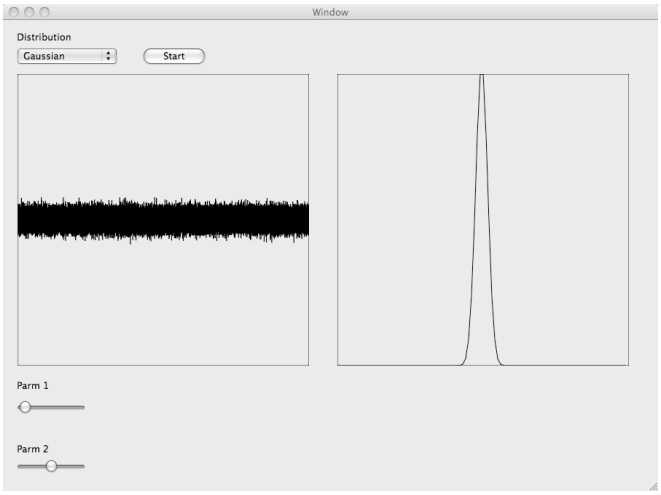


Figure 3. Output of the Gaussian generator, with its probability distribution for 100,000 samples. The mean is still 0.5, but the deviation is reduced, producing a series of outputs that are closer to the mean.

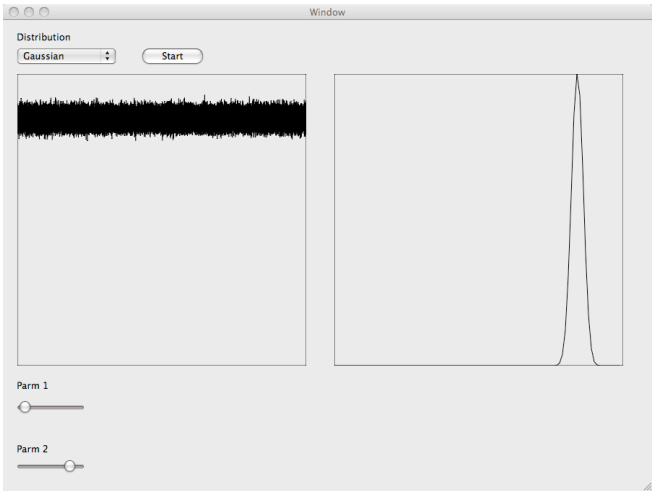


Figure 4. Output of the Gaussian generator, with its probability distribution for 100,000 samples. The mean has been raised, while the deviation has been kept the same as figure 3. Now the outputs are clustered around the new (higher) mean.

In order to perform with "Probability Jammer", the composer needs to specify the type of probability distributions to be applied to pitch and duration, select the synthesizer (all specified using the ChucK language), and initialise their controlling parameters. Various probability distributions, taken from the standard literature [2,3,5], can be applied to the pitch and duration, such as Gaussian, Linear, Exponential and Triangle probability distributions (Settings for using Gaussian are shown in figure 1). One other improvisation was made with another ChucK instrument that is a variation of "Probability Jammer". This instrument is optimized for making multilayered

drone like textures. It makes use of a Gaussian probability distribution to choose pitches that are at harmonic frequencies of a note specified by the user, via a keyboard, using a number of STK synthesizers.

Different strategies were employed by the composer to create the different improvisations that form the raw material for this piece. In a few improvisations, there were no changes in the controls during the recording phase. Once the instrument settings were producing a distinctive output that the composer found interesting, the audio was recorded for around thirty seconds. One example of this approach is the improvisation used at the very beginning of the piece. It makes use of a Gaussian distribution to control the pitch, with the mean pitch set quite low and a medium deviation around this pitch. The duration is set so as to produce a constant, fast, note rate. As these parameters do not change, all of the notes generated exhibit these characteristics. Most of the improvisations, such as the solo melodic line in the middle of the piece, were created by initializing the system to a configuration that the composer felt was a good starting point, and then performing the improvisation by adjusting the probability distribution parameters while the part was being recorded. In this situation, the composer was usually thinking in terms of melodic line, aiming at creating a recording with a degree of variation in the note rate and the pitch. There was an attempt to create a distinct pitch contour, with some rhythmic variation.

2.2. Phase 2 – Processing and Editing

The second phase of the composition process involves "dressing up" the timbre of the raw improvisations (which sound very raw indeed) to create a basic palate with which to assemble the piece. The recordings of the ChucK improvisations were imported into Logic Pro8 and signal processing was applied to each part so as to accentuate an aspect of its character that appealed to the composer. The aim was to create a collection of distinct musical "gestures" that would be recognizable, even after undergoing various transformations, as these will be the elements that provide coherence and can help a listener make sense of the piece.

All of the standard modern signals processing options, such as equalisation, reverb, compression etc, were utilized as seemed appropriate, during this phase of production. It is worth noting that a large part of the characteristic surface "sound" of the piece came from extensive use of the "EnVerb" plug in (a combination of a reverb and an envelope shaper), the Rotary Speaker simulator, as well as various types of delay and distortion. Offline Pitch Shifting was applied to some of the audio files, to create some of the low pitched parts. Additionally, in order to enhance the "sound cloud" effect that was apparent in some of the improvisations, those recordings were duplicated, cut into sections and reordered, and layered with some timing offsets. This resulted in a more diffuse cloud of notes with the same overall pitch characteristics.

2.3. Phase 3 – Assembly and Mix

Once the palate of parts was in place, the piece was assembled. This took place over a period of one month. It was done using a trial and error process involving experimenting with different combinations of parts so as to create a distinct character that appealed to the composer. It was important to the composer that the final form of the piece exhibited a certain clarity/transparency in its overall sound. To help achieve this, experiments creating different textural layers, with various degrees of timbral contrast, were undertaken. After some time, nine distinct textural layers and four distinct sections emerged:

1. High and low organ like flourishes over the multilayer drone.
2. Swirling "cloud like" textures.
3. A melodic section making use of the "bowed" STK instrument.
4. A noise/percussion section.

In this phase of the construction, the texture and timbre of large groups of notes were the main concern, rather than the details of the individual notes. Pitch is only organized in terms of predominant register (High, Medium, Low).

The primary concern was to achieve a sense of cohesion and progression through the duration of the piece, with a convincing musical continuity (with appropriate balance between "flow" and "break"). In its final form, the piece starts quite lively and progress towards an (anti) climax, where it virtually stops, and then returns to some state similar to the beginning (a variation of a traditional ternary form). The final form is clearly seen in the screen shot of the arrange page of Logic Pro, shown in figure 5.

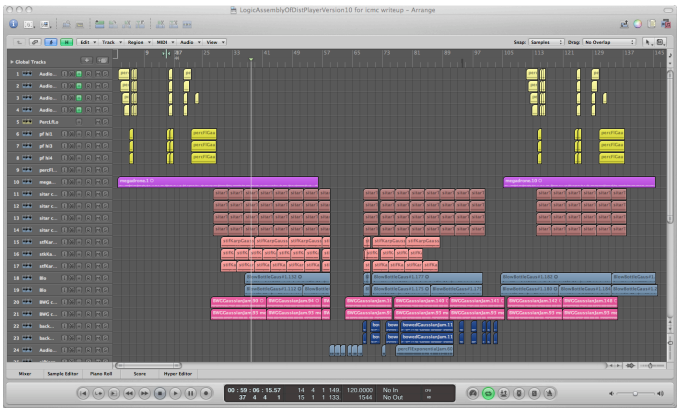


Figure 5. Logic Arrange page, indicating the overall structure of the piece.

3. HISTORICAL CONTEXT

"In Transit" is part of a project that attempts to apply, in an electro acoustic context, some of the processes that the composer utilized working with rock/folk/jazz groups (and recording sessions for TV and film) in the 1980s. Central to this approach was that players made use of distinct strategies to come up with their own part, rather than playing a set "arrangement". The arrangement emerged spontaneously, in real time. In that situation, the strategies were often "rules of thumb" for creating common textural elements. An example would be the various different approaches used to create the idiomatic rock/jazz/ballade/country bass lines. Approaches for these could be (amongst many others):

- rock - play the root of the chord in quavers.
- jazz - play a "walking bass" making sure to clearly emphasise the chord tones.
- ballade - play the root of the chord dotted crotchet-quaver pattern.
- country – play the root and fifth of the chord in minimums.

Similar idiomatic approaches were applied to the creation of "comping" patterns, pads, arpeggio parts and counter melodies etc. Often, this approach is moderated (often by a producer) either in rehearsal, while the parts are being recorded (especially in a multitrack recording situation) or after the performance was recorded. Mixing, signal processing and editing techniques, such as copying/cutting and pasting/splicing, have been utilized extensively to modify recorded improvised performances, since the mid 1960s. A well-known example is "Bitches Brew" by Miles Davis, where the raw recordings of the musicians were heavily edited to create the final record. (Ironically, for this project, "Bitches Brew" was influenced by the work of the Musique Concrete composers.)

The process used to create "In Transit" is an obvious descendent of the above process. The "rules of thumb" are replaced by the distinct improvisation strategies used to create performances with the "Probability Jammer" (directly inspired by Xenakis), and the finished composition emerges through choices made by the composer in the assembly process, making use of extensive studio production techniques.

4. CONCLUSION

In this piece, the composer has made use of a compositional process that attempts to create an electro acoustic piece that combines the elements the spontaneity of an improvised performance with the careful consideration of a pre-composed piece. The process provides a lot of opportunities to create interesting musical results through the serendipitous juxtaposition and superposition of material. The approach also enables the possibility of musical decision making skills developed in other musical genres to be applied in an electro acoustic context, which may be considered, by some people, as a desirable attribute.

5. REFERENCES

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